# Cielo Epic AIoT Platform Architecture

Scalable MQTT Infrastructure using RabbitMQ for ESG, Energy & Smart Infrastructure

INSPIRONICS CORPORATION

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# Unified AIoT Backbone for Cielo Epic

Cielo Epic is engineered to serve as the single, unified intelligence layer connecting devices, applications, and analytics across distributed infrastructure. It is purpose-built for organizations demanding real-time data ingestion, ESG accountability, and enterprise-scale deployment.

### Single Platform

Connects devices, apps, and analytics in one unified backbone

### Real-Time Ingestion

Continuous data flow from distributed infrastructure at scale

### ESG-Native

Automated insights and policy enforcement for sustainability goals

### $100M-Scale

Multi-tenant SaaS architecture designed for enterprise deployment

# Fragmented IoT & Data Silos

Today's infrastructure suffers from disconnected systems, reactive operations, and inadequate security. Without a unified messaging layer, enterprises cannot act on real-time data, enforce ESG mandates, or scale across geographies efficiently.

### Device Isolation

Sensors and edge devices operate in silos, preventing cross-system intelligence and coordinated automation responses.

### No Real-Time Intelligence

Batch-based pipelines introduce latency, making anomaly detection, alerts, and dynamic decisions impractical at scale.

### Poor Geo-Scalability

Proprietary protocols and regional brokers create bottlenecks and operational complexity when expanding across sites.

### Security & Identity Gaps

IP-based authentication and weak identity frameworks expose enterprises to spoofing, data breaches, and compliance risk.

### Inefficient ESG Monitoring

Siloed energy and emissions data prevents accurate carbon accounting, regulatory compliance, and sustainability optimization.

# MQTT-Based Unified Messaging Layer

A central RabbitMQ-powered MQTT broker acts as the intelligent backbone, enabling all devices, services, and applications to communicate through a single, governed, high-throughput channel.

SOLUTION OVERVIEW

## 1 Central MQTT Broker

RabbitMQ cluster handles millions of concurrent connections with QoS guarantees and durable message routing.

## 2 Publish–Subscribe Architecture

Real-time, decoupled messaging eliminates polling and enables reactive, event-driven workflows.

## 3 Multi-Region Connectivity

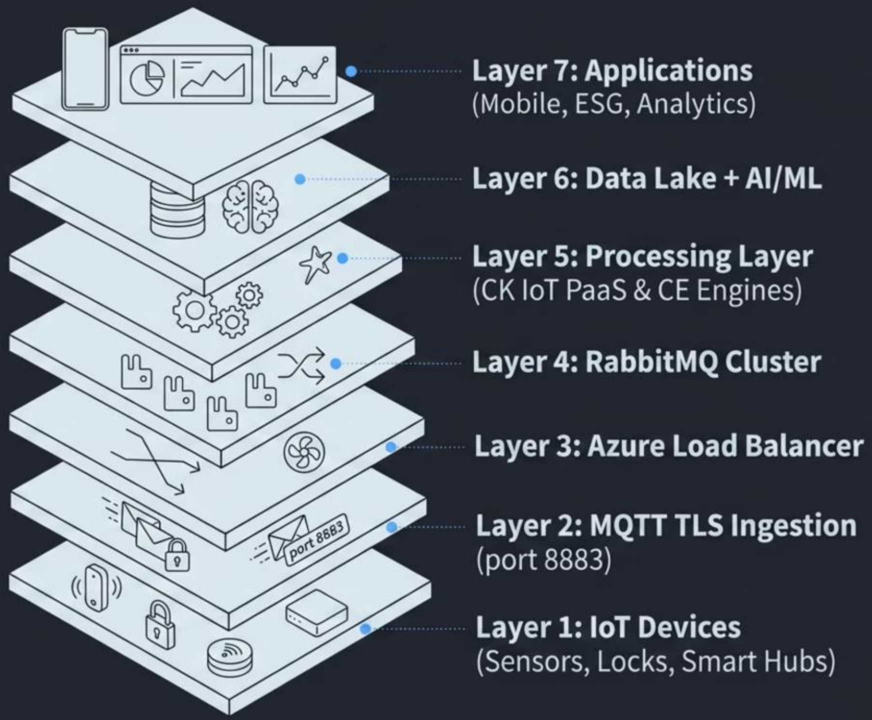
Federated broker topology connects devices and consumers across geographies with low latency.

## 4 Secure & Extensible

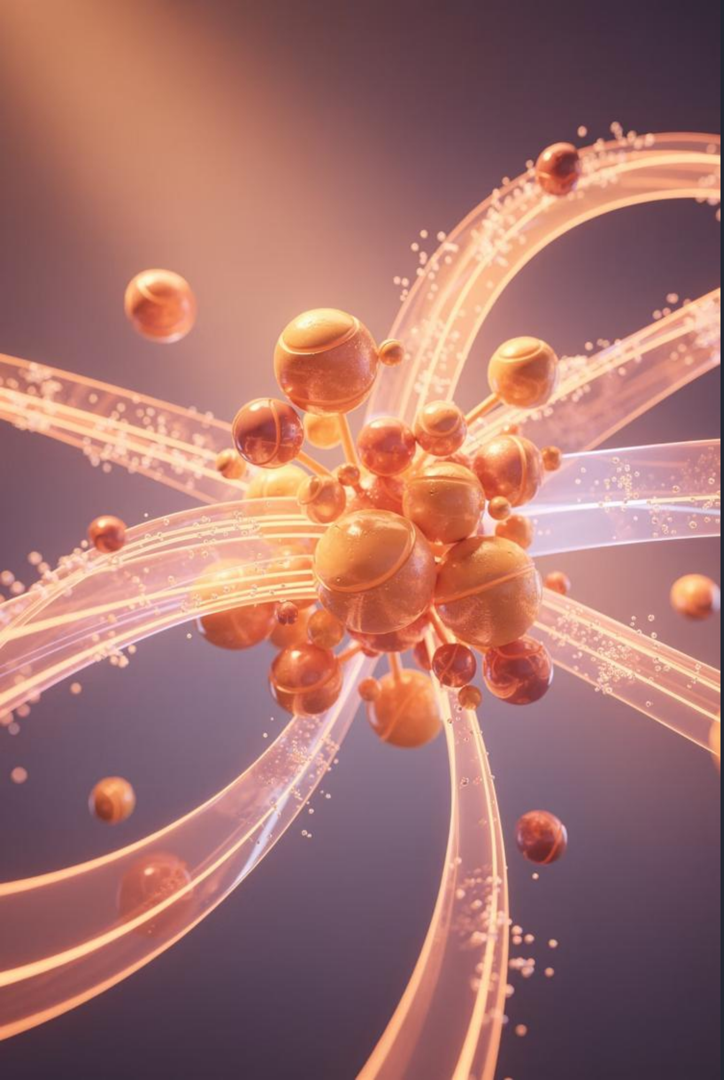
TLS encryption, certificate-based identity, and plugin architecture future-proof the platform.

# End-to-End Platform Flow

The Cielo Epic architecture is organized into seven distinct functional layers, each responsible for a specific stage of data movement, processing, and application delivery.




Each layer is independently scalable, enabling horizontal growth without architectural disruption. From edge sensors through to end-user applications, every data point is brokered, secured, and actionable.



# Publish–Subscribe Model

The MQTT publish–subscribe model is the cornerstone of Cielo Epic's real-time intelligence. Devices act as publishers, continuously emitting telemetry without needing awareness of downstream consumers. This decoupling enables massive parallelism and elastic subscriber scaling.

## Device Publishing

Edge devices publish telemetry, alerts, and status to structured topic namespaces continuously and autonomously.

## Multi-Consumer Delivery

Multiple subscribers consume the same topic simultaneously — dashboards, AI engines, and mobile apps in parallel.

## Real-Time Automation

Event-driven subscriptions trigger instant actions — alerts, HVAC adjustments, security responses — without polling delays.

# Unlimited Subscriber Scalability

A defining strength of the Cielo Epic MQTT architecture is its ability to fan out a single topic to an unlimited number of concurrent subscribers — each receiving the full message stream in real time, without additional publisher-side overhead.

### Multiple Subscriber IPs

Broker routes identical topics to any number of independent consumer endpoints across networks and regions.

### Parallel Application Delivery

Dashboards, AI inference engines, mobile apps, and logging systems all consume simultaneously with zero contention.

### Shared Subscriptions

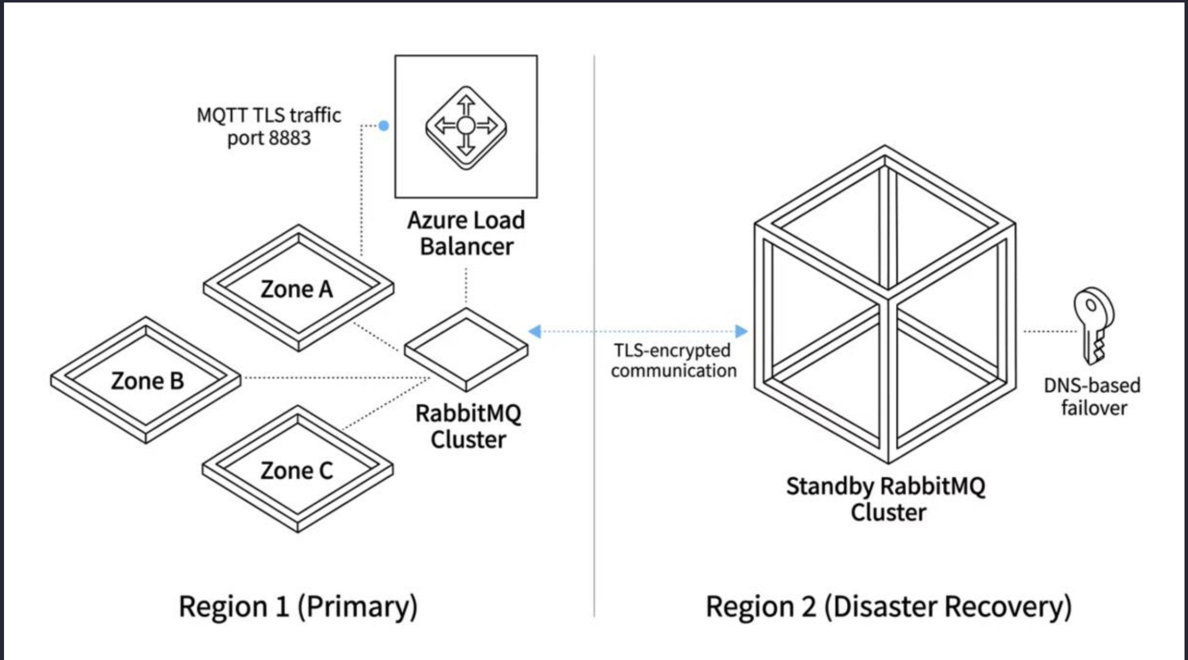
Optional load-balanced shared subscriptions distribute message load across consumer instances for throughput scaling.

### Elastic Consumer Pools

Subscriber capacity can be scaled independently of broker infrastructure, enabling cost-efficient compute allocation.

# Production-Grade Azure Deployment

The Cielo Epic platform is deployed on Microsoft Azure with a multi-region, high-availability topology. Primary and disaster recovery clusters are geographically separated, ensuring business continuity even during regional outages.




### 3-Node Primary Cluster

Quorum-based consensus ensures no single point of failure within the primary region.

### Azure Load Balancer

Distributes MQTT connections with health probing and automatic rerouting on node failure.

### DR Region

Geo-redundant standby cluster activated via DNS failover within recovery time objectives.

### TLS Port 8883

All device-to-broker communication encrypted end-to-end; no plaintext MQTT permitted.

# Zero-Trust Device Security

Cielo Epic's security model eliminates implicit trust at every layer. Devices are authenticated by cryptographic identity, not network location, ensuring that no actor — inside or outside the network perimeter — gains access without explicit, verifiable credentials.

### Certificate-Based Identity

Each device authenticates via a unique `client_id` and X.509 certificate — no shared secrets.

### End-to-End TLS

All MQTT traffic is encrypted in transit using TLS 1.2/1.3, preventing interception and tampering.

### Topic-Level ACLs

Granular publish and subscribe permissions enforce least-privilege access per device and per tenant.

### Virtual Host Isolation

Each tenant operates within a dedicated RabbitMQ virtual host, ensuring complete data and namespace segregation.

# Structured Topic Namespace Design

A well-designed topic hierarchy is foundational to scalable IoT governance. The Cielo Epic namespace encodes context — tenant, site, building, device type, and device ID — directly into every topic path, enabling precise routing, security enforcement, and analytics segmentation without additional metadata lookups.

**Telemetry:** `cielo/{tenant}/{site}/{building}/{deviceType}/{deviceId}/telemetry`
**Alerts:** `cielo/{tenant}/{site}/{deviceId}/alerts`
**Commands:** `cielo/{tenant}/{site}/{deviceId}/commands`

### Clean Routing

Wildcard subscriptions on tenant or site segments allow selective fan-out without additional broker configuration.

### Analytics Segmentation

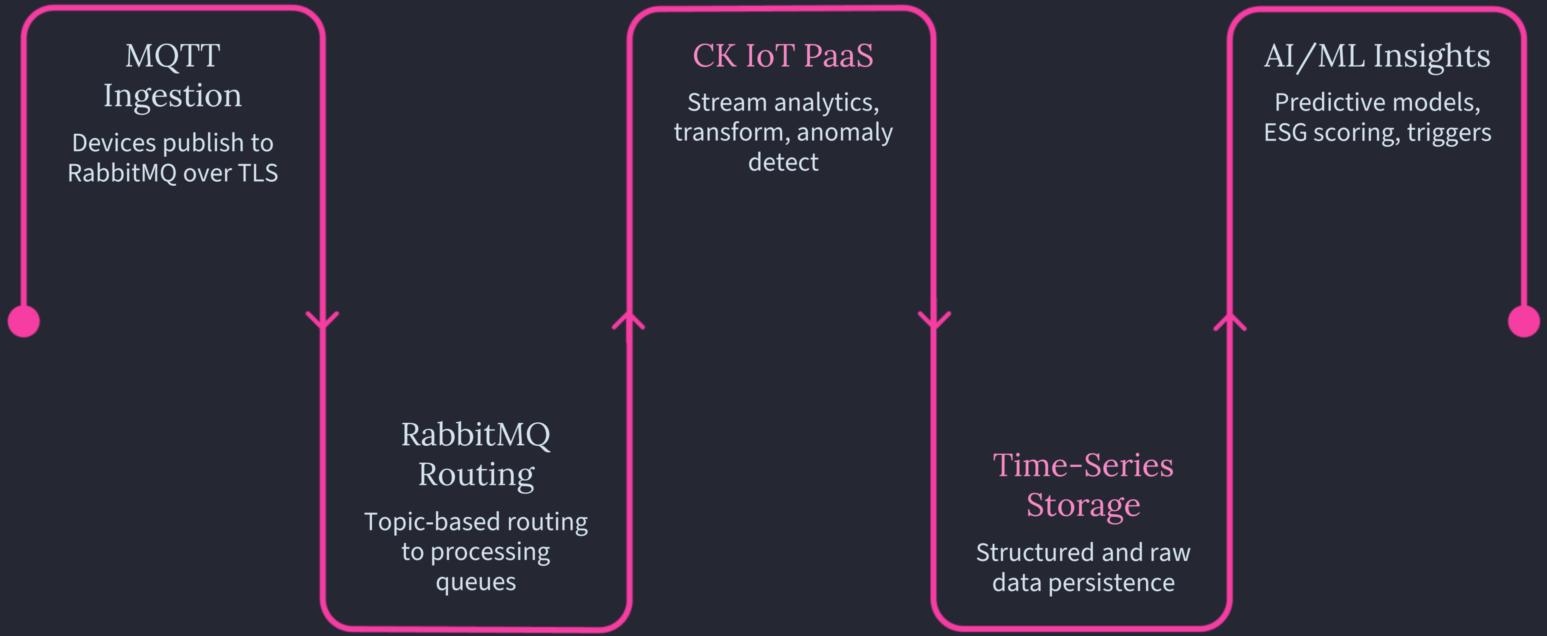
Downstream consumers filter by site, building, or device type using topic patterns, enabling precise data aggregation.

### Governance & Audit

Structured paths make access control policies, audit logs, and data lineage straightforward to implement and verify.

# Real-Time Data Pipeline

From edge ingestion to AI-driven insight, the Cielo Epic data pipeline is engineered for sub-second latency and continuous processing. Every message traverses a governed, observable path with transformation, enrichment, and storage at each stage.




The pipeline supports both real-time stream processing for immediate alerting and batch analytics for trend analysis, ESG reporting, and model training — all from a single, unified message flow.

# Platform Convergence: Cielo Epic Stack

Cielo Epic does not operate in isolation — it is the convergence point for a comprehensive suite of purpose-built platforms and applications. Each component of the Inspironics ecosystem plays a defined role in the end-to-end data and user journey.

### Caleido Kombos

The primary IoT ingestion layer; aggregates sensor data and publishes structured telemetry into the MQTT backbone.

### Cielo Epic

The ESG AI platform; consumes processed data streams to deliver carbon accounting, energy optimization, and compliance dashboards.

### Caleido Domi

The sensor ecosystem layer providing environmental, occupancy, and energy monitoring across buildings and campuses.

### Mobile Apps — Porta, Cherisimio, Tirisi

Guest-facing and operator-facing mobile interfaces that subscribe to real-time device and event streams for dynamic UX delivery.

### Mints

Loyalty and wellness integration layer; consumes occupancy and behavioral data to deliver personalized, context-aware experiences.

# Horizontal Scale Strategy

Cielo Epic is designed to grow without architectural rewrites. Every layer of the stack — brokers, consumers, storage, and compute — can be scaled independently and horizontally, enabling linear throughput growth as device fleets and data volumes expand.

### Broker Node Expansion

Add RabbitMQ nodes to the cluster to increase message throughput and connection capacity without downtime.

### Independent Subscriber Scaling

Consumer services scale on dedicated compute, allowing AI engines and dashboards to grow on their own schedules.

### Multi-Region Expansion

New geographic regions are onboarded by provisioning federated broker clusters — no changes to device firmware required.

### Cloud-Native Elasticity

Azure-native auto-scaling policies adjust compute and storage resources dynamically based on real-time load metrics.

## 100M+

Messages / Day

Target throughput at full enterprise scale deployment

## 3+

Broker Nodes

Minimum cluster size, expandable linearly

## ∞

Subscribers

No architectural limit on concurrent topic consumers

# Real-World Applications

The Cielo Epic MQTT platform is purpose-built for industries where real-time data, sustainability mandates, and operational intelligence converge. Each use case leverages the same unified backbone — reducing integration cost while maximizing domain-specific value.

### Smart Hospitality

Real-time room energy management, keyless access control, and guest experience personalization via connected devices and mobile apps.

### Smart Buildings

HVAC automation, occupancy-based lighting, and predictive maintenance driven by continuous sensor telemetry and AI analytics.

### ESG Monitoring

Real-time carbon tracking, energy consumption benchmarking, and automated ESG compliance reporting across portfolio assets.

### Industrial IoT

Predictive maintenance for industrial equipment using anomaly detection on continuous vibration, temperature, and pressure telemetry.

### Smart Cities

Municipal infrastructure management including traffic flow, public utility monitoring, and citizen-facing services powered by edge intelligence.

# Enterprise-Grade Reliability & Resilience

Mission-critical infrastructure demands zero tolerance for message loss and minimal recovery time. Cielo Epic's resilience architecture layers redundancy at every tier — from individual broker nodes to cross-region failover — ensuring continuous availability under both planned and unplanned failure conditions.

### Clustered RabbitMQ

Multi-node quorum queues prevent data loss on individual node failure; consensus-based writes ensure message durability.

### Zone Redundancy

Broker nodes are distributed across Azure Availability Zones, eliminating data center-level single points of failure.

### DR Region

A geographically separated disaster recovery cluster replicates critical queues and activates via DNS failover within defined RTO/RPO windows.

### Message Durability

Persistent message options ensure no telemetry is lost during transient consumer downtime or processing lag spikes.

### DNS-Based Failover

Automated DNS health checks reroute MQTT traffic to the DR cluster within seconds, transparent to all connected devices.

# Why This Architecture Wins

A purpose-built AIoT backbone that is simultaneously real-time, multi-tenant, ESG-native, and acquisition-ready — a combination no incumbent platform currently delivers at this scale.

### True Multi-Tenancy

Virtual host isolation and topic-level ACLs deliver genuine tenant separation — not just logical segmentation — at the broker level.

### Real-Time at Scale

Sub-second message delivery from edge to application, supporting millions of concurrent connections without degradation.

### ESG-Native Design

Sustainability metrics are first-class data citizens — not afterthoughts — enabling automated compliance and carbon reporting.

### Seamless Integration

Open MQTT standards ensure compatibility with any device, cloud, or analytics platform — past, present, and future.

### M&A Ready

A clean, documented, broker-centric architecture is highly attractive to strategic acquirers and technology partners evaluating platform value.

# Execution Roadmap

Cielo Epic follows a phased delivery model that validates core capabilities early while systematically expanding toward full global deployment and AI-driven monetization. Each phase builds on the previous, reducing risk and accelerating time-to-value.

### 1 Phase 1 — MVP

Single-region deployment with a 3-node RabbitMQ cluster. Core MQTT ingestion, TLS security, and basic topic namespace validated with initial device fleet.

### 2 Phase 2 — Pilot

Live deployments in Smart Hospitality and ESG monitoring verticals. CK IoT PaaS processing layer activated; mobile apps connected to live telemetry streams.

### 3 Phase 3 — Expansion

Multi-region broker federation across primary and DR Azure regions. Subscriber pool scaling, shared subscriptions, and full Cielo Epic stack integration.

### 4 Phase 4 — Monetization

AI-driven automation, predictive ESG scoring, and platform-as-a-service monetization. Strategic partner and acquirer enablement with full documentation and SLAs.

# Architecture at a Glance

A consolidated view of key technical parameters and design choices that define the Cielo Epic MQTT platform for stakeholder reference and due diligence evaluation.

| Dimension | Specification | Rationale |
|---|---|---|
| Broker | RabbitMQ Cluster (3+ nodes) | Quorum queues, plugin ecosystem, MQTT 5.0 support |
| Protocol | MQTT over TLS (port 8883) | Lightweight, industry-standard IoT messaging |
| Auth Model | X.509 Certificates + ACLs | Zero-trust, IP-independent device identity |
| Tenancy | Virtual Host Isolation | Hard namespace and data separation per tenant |
| Deployment | Azure Multi-Region | HA with Availability Zones + DR failover |
| Processing | CK IoT PaaS + CE Engines | Real-time stream analytics and anomaly detection |
| Storage | Time-Series DB + Data Lake | Optimized for telemetry query and ML model training |
| Scale Target | 100M+ messages/day | Supports enterprise-scale device fleets globally |

# Key Risks & Mitigations

Proactive risk management is integral to the Cielo Epic delivery strategy. The following risks have been identified, assessed, and addressed with specific architectural or operational mitigations.

### Broker Overload at Scale

**Risk:** Message volume exceeds single-cluster capacity. **Mitigation:** Horizontal node addition and sharding via consistent hash exchange routing distributes load without topology changes.

### Certificate Lifecycle Management

**Risk:** Expired device certificates cause authentication failures at scale. **Mitigation:** Automated certificate rotation integrated with Azure Key Vault and device management APIs.

### Cross-Region Latency

**Risk:** Federation lag introduces data consistency issues. **Mitigation:** Async replication with region-local processing ensures applications are never blocked by inter-region propagation.

### Tenant Data Leakage

**Risk:** Misconfigured ACLs expose cross-tenant data. **Mitigation:** Virtual host isolation as primary boundary; automated ACL validation in CI/CD pipeline prevents misconfigurations from reaching production.

# The Future of Intelligent Infrastructure

## One Platform

A single, unified MQTT backbone serving every device, application, and insight across the entire enterprise portfolio

## Infinite Devices

An architecture with no artificial ceiling — scale from hundreds to millions of connected endpoints without redesign

## Real-Time Intelligence

Sub-second telemetry, instant alerting, and continuous AI inference delivered from edge to executive dashboard

## ESG-Driven Outcomes

Sustainability is embedded in the architecture — every data point contributes to carbon accountability and compliance

*"Cielo Epic powers the Future of Cognitive Intelligence for every Infrastructure."*
— **Inspironics Corporation**

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